

The Linear Regression Model

Part III — Statistical Inference & Model Validation

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Statistical Inference in Regression

Lecture Overview

Lecture 10 — Statistical Inference & Model Validation

- Distributions of OLS estimators under normality
- t -tests for individual coefficients
- Confidence intervals for β_j
- The F -test: overall model significance
- The ANOVA table for regression
- Point prediction and prediction intervals
- Validating the classical assumptions:
 - Linearity, normality, homoscedasticity, independence, multicollinearity

Sampling Distributions of OLS Estimators

From Estimates to Inference

We have $\hat{\beta}_j$ — point estimates. But **how uncertain** are they?

Under the classical assumptions A1–A6:

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma_{\hat{\beta}_j}^2\right)$$

Since σ^2 is unknown, we replace it with $s^2 = \text{SSR}/(n - k - 1)$:

$$T_j = \frac{\hat{\beta}_j - \beta_j}{s_{\hat{\beta}_j}} \sim t_{n-k-1}$$

where $s_{\hat{\beta}_j}$ is the **standard error** of $\hat{\beta}_j$ (reported by R in the **Std. Error** column).

t-Tests for Individual Coefficients

The Individual Significance Test

Question: Is the variable X_j statistically significant — i.e. does it contribute to explaining Y ?

Hypotheses:

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j \neq 0$$

Test statistic:

$$t_j = \frac{\hat{\beta}_j}{s_{\hat{\beta}_j}} \sim t_{n-k-1} \quad \text{under } H_0$$

Decision rule at significance level α :

Reading the R Output

```
1 summary(mlr_model)
```

Call:

```
lm(formula = profit2 ~ revenue2 + employees)
```

Residuals:

Min	1Q	Median	3Q	Max
-21.8947	-9.0708	-0.1219	6.0029	25.3186

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	21.46125	9.51476	2.256	0.0324	*
revenue2	0.12902	0.02048	6.299	9.66e-07	***
employees	0.61262	0.11789	5.196	1.80e-05	***

Interpreting the t -test Output

From the `Coefficients` table in `summary()`:

Column	Meaning
Estimate	$\hat{\beta}_j$ — the OLS point estimate
Std. Error	$s_{\hat{\beta}_j}$ — standard error of the estimate
t value	$t_j = \hat{\beta}_j / s_{\hat{\beta}_j}$ — the test statistic
Pr(> t)	Two-sided p -value: probability of observing $ t_j $ this large if H_0 were true
Significance codes	*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$

⚠ Important

If $\text{Pr}(>|t|) < \alpha$ (e.g. 0.05): **reject** H_0 — the variable is statistically significant at level α .

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If not: we **fail to reject** H_0 — the variable may not be contributing to the model after controlling for the others

One-Sided Tests

Sometimes theory suggests a **directional** hypothesis:

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j > 0 \quad (\text{or } H_1 : \beta_j < 0)$$

The test statistic is the same $t_j = \hat{\beta}_j / s_{\hat{\beta}_j}$.

For $H_1 : \beta_j > 0$: reject H_0 if $t_j > t_{n-k-1, \alpha}$

For $H_1 : \beta_j < 0$: reject H_0 if $t_j < -t_{n-k-1, \alpha}$

Note

The p -value from R is always **two-sided**. For a one-sided test, divide it by 2 — but only when the sign of $\hat{\beta}_j$ is consistent with H_1 .

Confidence Intervals for β_j

Construction

A $(1 - \alpha) \times 100\%$ confidence interval for β_j :

$$\hat{\beta}_j \pm t_{n-k-1, \alpha/2} \cdot s_{\hat{\beta}_j}$$

! Important

Interpretation: If we repeated the sampling procedure many times, $(1 - \alpha) \times 100\%$ of the intervals constructed this way would contain the true β_j .

The interval gives a range of **plausible values** for β_j given the data.

Link to hypothesis testing: if 0 is **not** in the 95% confidence interval for β_j , then $H_0 : \beta_j = 0$ is rejected at $\alpha = 5\%$.

Confidence Intervals in R

```
1 confint(mlr_model, level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	1.93857748	40.9839143
revenue2	0.08699515	0.1710521
employees	0.37072535	0.8545246

Interpretation:

- The 95% CI for β_{revenue} is approximately [0.087, 0.171].
- Since 0 is **not** in this interval, we reject $H_0 : \beta_{\text{revenue}} = 0$ at 5% — revenue has a statistically significant positive effect on profit.

The F -Test: Overall Significance

Motivation

The t -tests assess **individual** regressors. But sometimes we want to ask:

Does the model as a whole explain a significant amount of variation in Y ?

This is the **overall significance test**:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0$$

! Important

Under H_0 , none of the regressors help explain Y — the model is no better than simply predicting $\bar{Y}$ for everyone.

The F -Statistic

$$F = \frac{\text{SSE}/k}{\text{SSR}/(n - k - 1)} = \frac{\text{Explained MS}}{\text{Residual MS}} \sim F_{k, n-k-1} \quad \text{under } H_0$$

The ANOVA table for regression:

Source	SS	df	MS	F
Regression (Explained)	SSE	k	SSE/k	MSE/MSR
Residual	SSR	$n - k - 1$	$\text{SSR}/(n - k - 1)$	
Total	SST	$n - 1$		

Decision: Reject H_0 if $F > F_{k, n-k-1, \alpha}$ or if $p\text{-value} < \alpha$.

F -Test in R

The F -statistic and its p -value appear at the **bottom** of `summary()`:

```
1 summary(mlr_model)
```

Call:

```
lm(formula = profit2 ~ revenue2 + employees)
```

Residuals:

Min	1Q	Median	3Q	Max
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Coefficients:

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The line **F-statistic: ... on k and n-k-1 DF, p-value: ...** gives the overall test.

Relationship Between F and R^2

In MLR, the F -statistic can be written as:

$$F = \frac{R^2 / k}{(1 - R^2) / (n - k - 1)}$$

This makes clear that:

- A high R^2 tends to produce a large F (and small p -value)
- But a significant F does **not** mean all individual coefficients are significant
- And a significant t_j does **not** imply overall model significance

Note

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In SLR ($k = 1$): $F = t^2$ — the F -test is equivalent to the two-sided t -test on the single slope

Prediction

Point Prediction

Given new values $X_1 = x_1^*, \dots, X_k = x_k^*$, the **point prediction** is:

$$\hat{Y}^* = \hat{\beta}_0 + \hat{\beta}_1 x_1^* + \dots + \hat{\beta}_k x_k^*$$

```
1 # Predict profit for revenue = 300, employees = 40
2 new_obs <- data.frame(revenue2 = 300, employees = 40)
3 predict(mlr_model, newdata = new_obs)
```

```
1
84.67332
```

⚠ Important

Extrapolation warning: predictions are only reliable within the range of the observed data. Predicting far outside the sample range (e.g. revenue = 5000 when the data only covers 100–500) is unreliable.

Prediction Intervals

A **prediction interval** accounts for two sources of uncertainty:

1. Uncertainty about the true mean $E[Y^*]$ (same as a confidence interval for the mean)
2. Uncertainty about the **individual** outcome Y^* around that mean

$$\hat{Y}^* \pm t_{n-k-1, \alpha/2} \cdot s \cdot \sqrt{1 + \mathbf{x}^{*'}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}^*}$$

In R:

```
1 predict(mlr_model, newdata = new_obs,
2         interval = "prediction", level = 0.95)
```

	fit	lwr	upr
1	84.67332	58.73202	110.6146

Validating the Classical Assumptions

Why Validation Matters

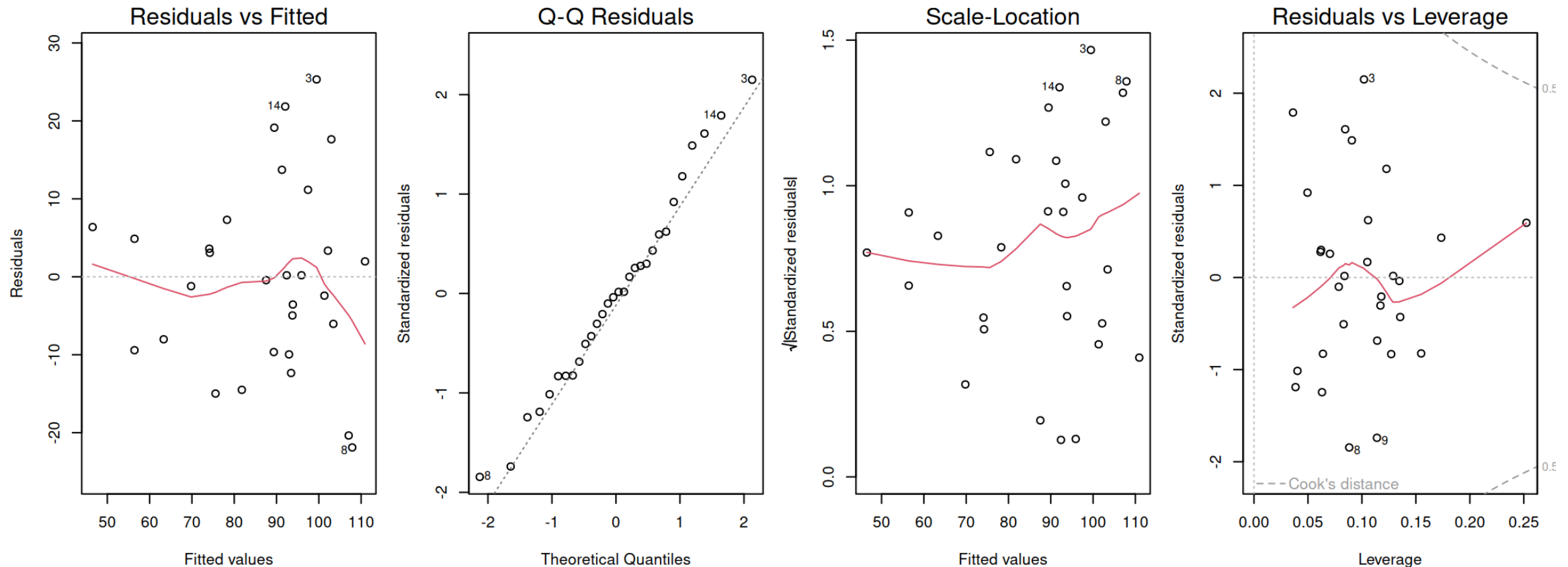
Our inference (t-tests, F-test, confidence intervals) is only valid if the classical assumptions hold.

If they fail:

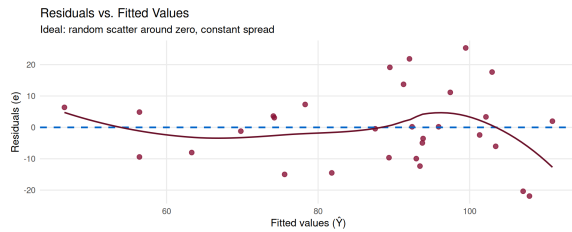
- **A2 violated** (non-zero mean error, omitted variables): OLS is biased
- **A3 violated** (heteroscedasticity): OLS standard errors are wrong → invalid t and F tests
- **A4 violated** (autocorrelation): same problem as heteroscedasticity
- **A6 violated** (non-normality): t and F approximations may fail in small samples
- **Multicollinearity** (not a classical assumption violation): inflated standard

The Four Key Diagnostic Plots

```
1 par(mfrow = c(1, 4), mar = c(4, 4, 3, 1))
2 plot(mlr_model)
```



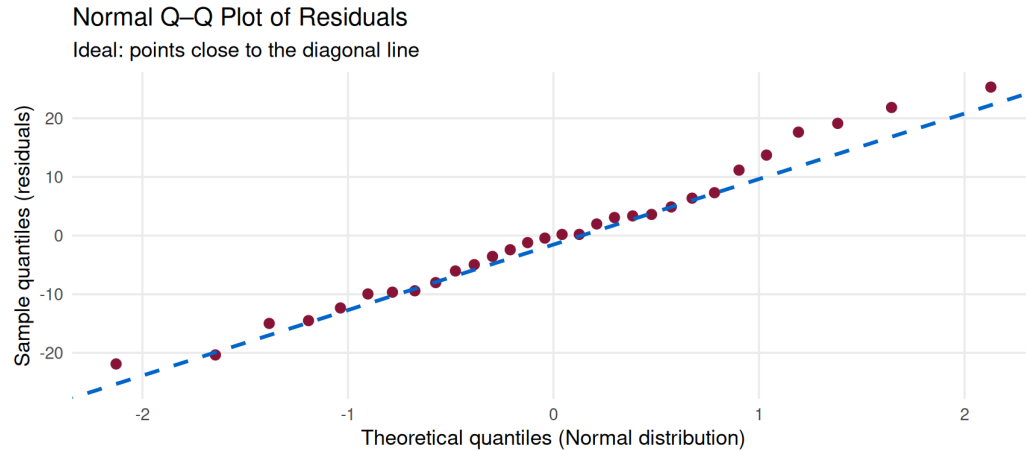
Residuals vs. Fitted: Checking Linearity and Homoscedasticity



What to look for:

- **Random scatter around zero** → linearity OK
- **Constant vertical spread** → homoscedasticity OK
- **Funnel or fan shape** → heteroscedasticity (problem)
- **Curved pattern** → non-linearity (problem)

Normality of Residuals: Q–Q Plot

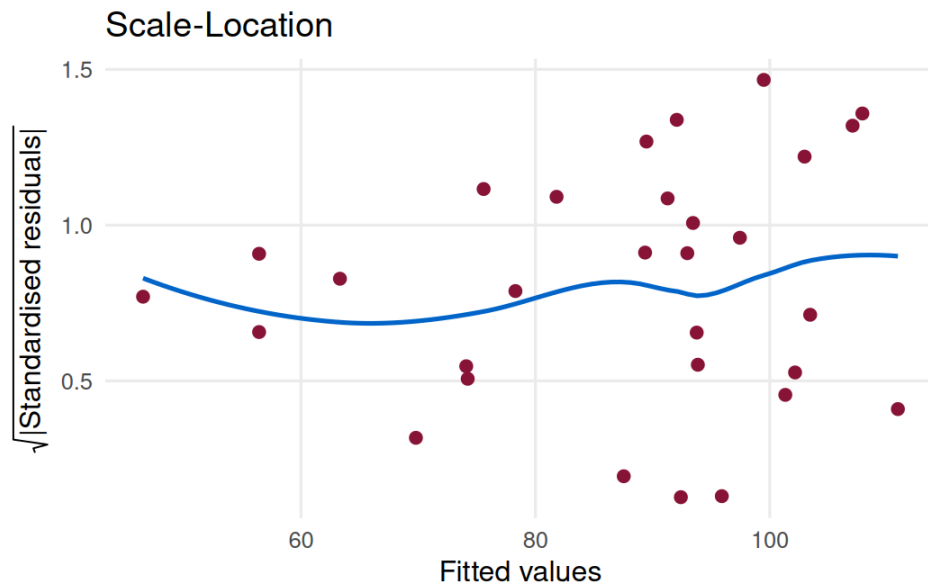


What to look for:

- **Points near the diagonal** → normality approximately satisfied
- **S-shaped curve** → heavy tails (non-normal)
- **Systematic departure** → consider transformations of Y

Detecting Heteroscedasticity

Scale-Location plot (square root of standardised residuals vs. fitted):



Ideal: the smooth line is **roughly horizontal** (constant spread).

Problem signs:

- Rising line → variance increases with $\hat{Y}$ (funnel-right)
- Falling line → variance decreases with $\hat{Y}$ (funnel-left)

Formal test: White's test (tests whether residual variance depends on **any regressor or its square**).

The Classical Assumptions: Summary Table

Assumption	How to check	Symptom if violated	Consequence
A1 Linearity	Scatter plot, residuals vs. fitted	Curved pattern	Biased estimates
A2 Zero mean	Residuals vs. fitted	Systematic non-zero residuals	Biased $\hat{\beta}_0$
A3 Homoscedasticity	Scale-location plot, White's test	Funnel shape	Invalid SEs & tests
A4 No autocorrelation	Durbin-Watson statistic, residuals vs. order	Pattern in residuals	Invalid SEs & tests
A6 Normality	Q–Q plot, Shapiro-Wilk test	S-curve in Q–Q	Invalid small-sample tests
—	Multicollinearity: VIF	High pairwise r , $VIF > 5$	Inflated SEs, unstable $\hat{\beta}_j$

Autocorrelation: The Durbin–Watson Test

Relevant mainly for **time series** data. Tests H_0 : no autocorrelation vs. H_1 : positive autocorrelation.

$$DW = \frac{\sum_{i=2}^n (e_i - e_{i-1})^2}{\sum_{i=1}^n e_i^2}$$

Rule of thumb: $DW \approx 2 \rightarrow$ no autocorrelation; $DW < 1.5 \rightarrow$ concern.

```
1 # install.packages("lmtest") if needed
2 library(lmtest)
3 dwtest(mlr_model)
```

Durbin-Watson test

```
data: mlr_model
DW = 1.6312, p-value = 0.1308
alternative hypothesis: true autocorrelation is greater than 0
```

Multicollinearity and VIF

Multicollinearity: two or more regressors are highly correlated with each other.

Consequence: OLS is still unbiased and BLUE, but standard errors are inflated
→ t -statistics are small → we may fail to detect significant effects.

The **Variance Inflation Factor (VIF)** for regressor X_j :

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 from regressing X_j on all other regressors.

Note

Rule of thumb: $\text{VIF} < 5$ is acceptable; $\text{VIF} > 10$ is a serious problem.

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VIF in R

```
1 # install.packages("car") if needed
2 library(car)
3 vif(mlr_model)
```

```
revenue2 employees
1.110802  1.110802
```

Both VIF values are close to 1 → no multicollinearity problem in our example.
This is expected since **revenue2** and **employees** were generated independently.

Putting It All Together: Model Validation Workflow

1. **Fit the model** with `lm()` and check `summary()` for coefficient signs, magnitudes and significance.
2. **Plot residuals vs. fitted** — check for linearity and homoscedasticity.
3. **Inspect the Q–Q plot** — check for normality of residuals.
4. **Run Durbin-Watson** (if data may have time ordering) — check for autocorrelation.
5. **Compute VIF** (if $k \geq 2$) — check for multicollinearity.
6. If violations are found: consider **transformations** ($\log Y$, $\log X$), **robust standard errors**, or **model re-specification** (add/remove variables).

Complete R Workflow

End-to-End: From Data to Validated Model

```
1 # 1. Fit the model
2 model <- lm(Y ~ X1 + X2, data = mydata)
3
4 # 2. Inspect results
5 summary(model)
6 confint(model, level = 0.95)
7
8 # 3. Residual diagnostics
9 par(mfrow = c(2, 2))
10 plot(model)
11
12 # 4. Autocorrelation (if needed)
13 library(lmtest)
14 dwtest(model)
15
16 # 5. Multicollinearity (if k >= 2)
17 library(car)
```

Reading a Regression Table

A typical regression output table (as seen in R or SPSS):

	Estimate	Std. Error	<i>t</i> value	Pr(> <i>t</i>)
(Intercept)	$\hat{\beta}_0$	$s_{\hat{\beta}_0}$	t_0	p_0
X_1	$\hat{\beta}_1$	$s_{\hat{\beta}_1}$	t_1	p_1
X_2	$\hat{\beta}_2$	$s_{\hat{\beta}_2}$	t_2	p_2

Metric	What it tells you
Residual std. error (s)	Typical size of residuals; in the same units as Y
R^2	Proportion of variance in Y explained
Adjusted R^2	R^2 penalised for number of regressors
F -statistic	Overall significance of the model

Course Summary: Linear Regression

The Linear Regression Journey (3 Lectures)

Lecture	Topic	Key Concepts
8	Introduction & OLS	Model specification, Gauss–Markov assumptions, OLS formulas, coefficient interpretation
9	Fit & MLR	R^2 , $\bar{R}^2$, Gauss–Markov theorem, Omitted Variable Bias, MLR <i>ceteris paribus</i> interpretation
10	Inference & Validation	t -tests, F -test, CIs, prediction intervals, residual diagnostics, VIF

The core message:

OLS gives us the best linear unbiased estimates — but only if the classical assumptions hold. Always validate your model before drawing conclusions.

Key Formulas at a Glance

	Formula
OLS slope	$\hat{\beta}_1 = S_{XY} / S_{XX}$
OLS intercept	$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$
Error variance	$s^2 = \text{SSR} / (n - k - 1)$
R^2	$1 - \text{SSR} / \text{SST}$
Adjusted R^2	$1 - (1 - R^2)(n - 1) / (n - k - 1)$
t -statistic	$t_j = \hat{\beta}_j / s_{\hat{\beta}_j}$
CI for β_j	$\hat{\beta}_j \pm t_{n-k-1, \alpha/2} \cdot s_{\hat{\beta}_j}$
F -statistic	$(\text{SSE} / k) / (\text{SSR} / (n - k - 1))$